

Scene Classification

Update 4

- Karina Damico
- Pauline Martial
- Piter Garcia

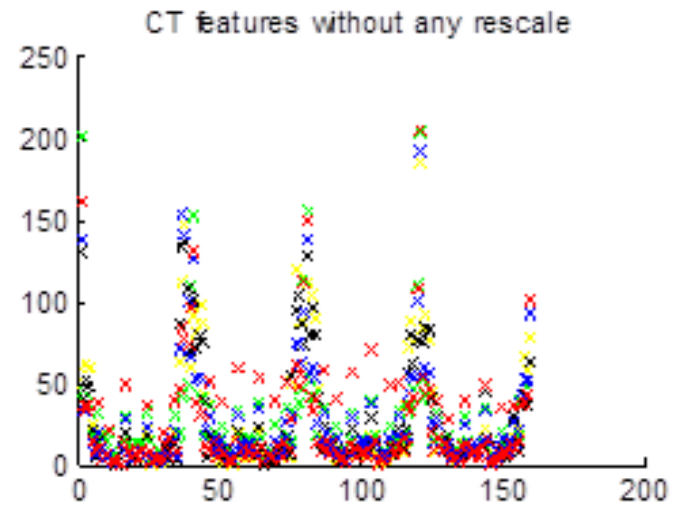
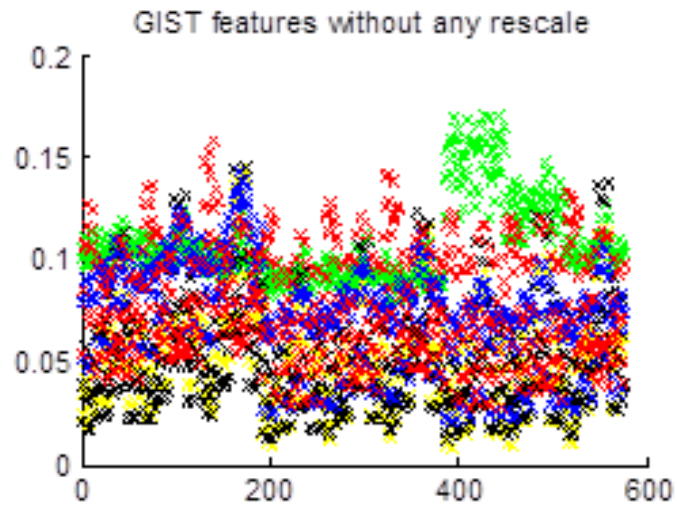
Karina and Pauline:
Gabor GIST + CENTRIST

Hybrid: Gabor GIST + CENTRIST

Visualize information of:

- GIST feature vectors
- CT feature vectors

Hybrid: Gabor GIST + CENTRIST



Very different scales!

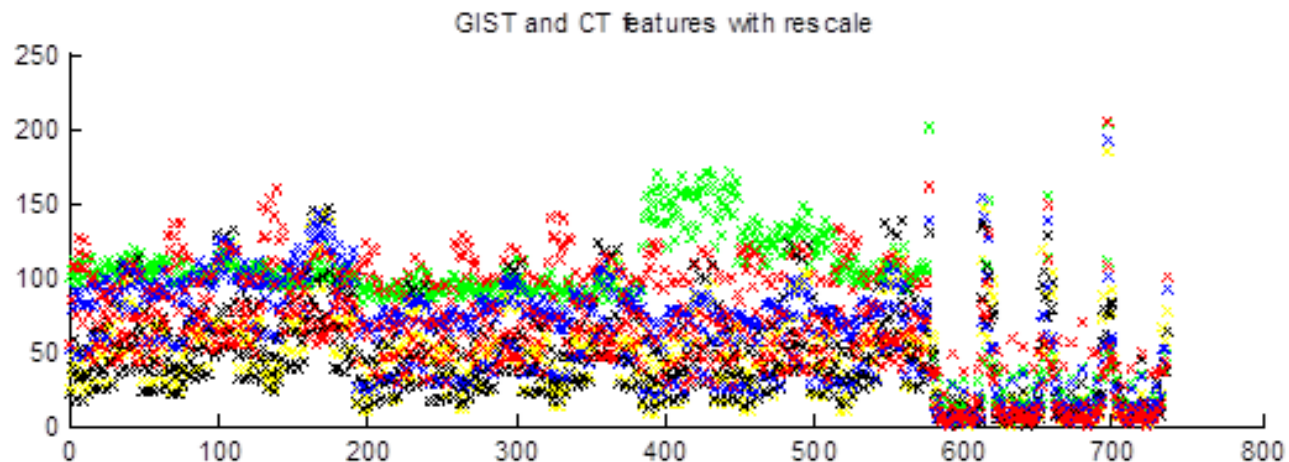
Hybrid: Gabor GIST + CENTRIST

How to scale?

Stretch Gabor GIST through CENTRIST scale

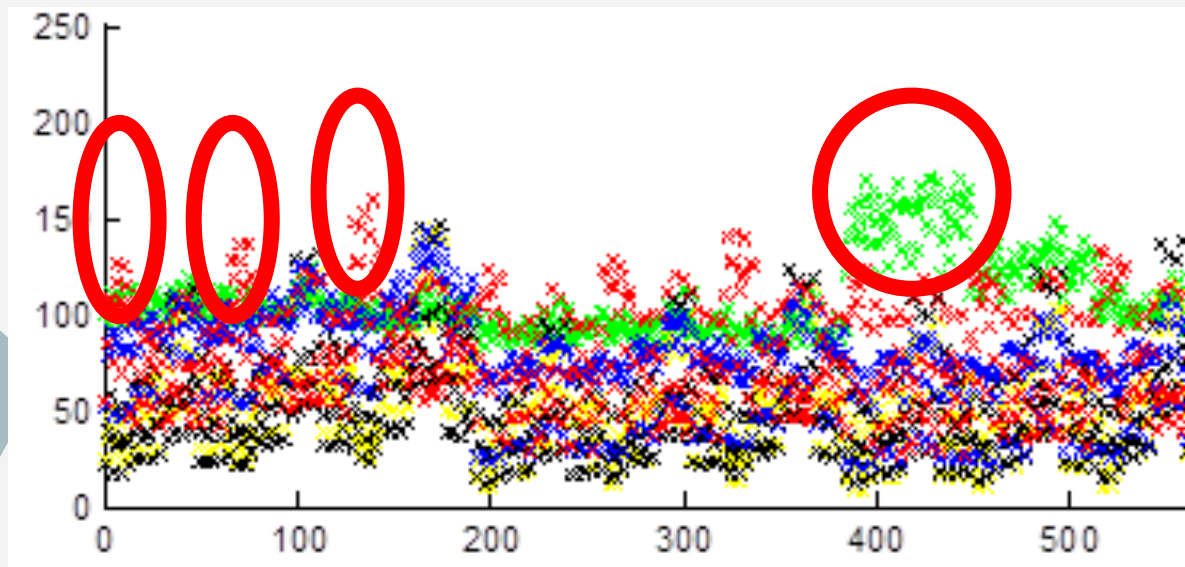
NN classifier rescales => absolute scale irrelevant
relative scale important

1. Find max and min of Centrist values: $[a, b]$
2. Scale Gabor into CENTRIST $[a, b]$ range: $A = (b-a)*\text{Gabor} + a$



Hybrid: Gabor GIST + CENTRIST

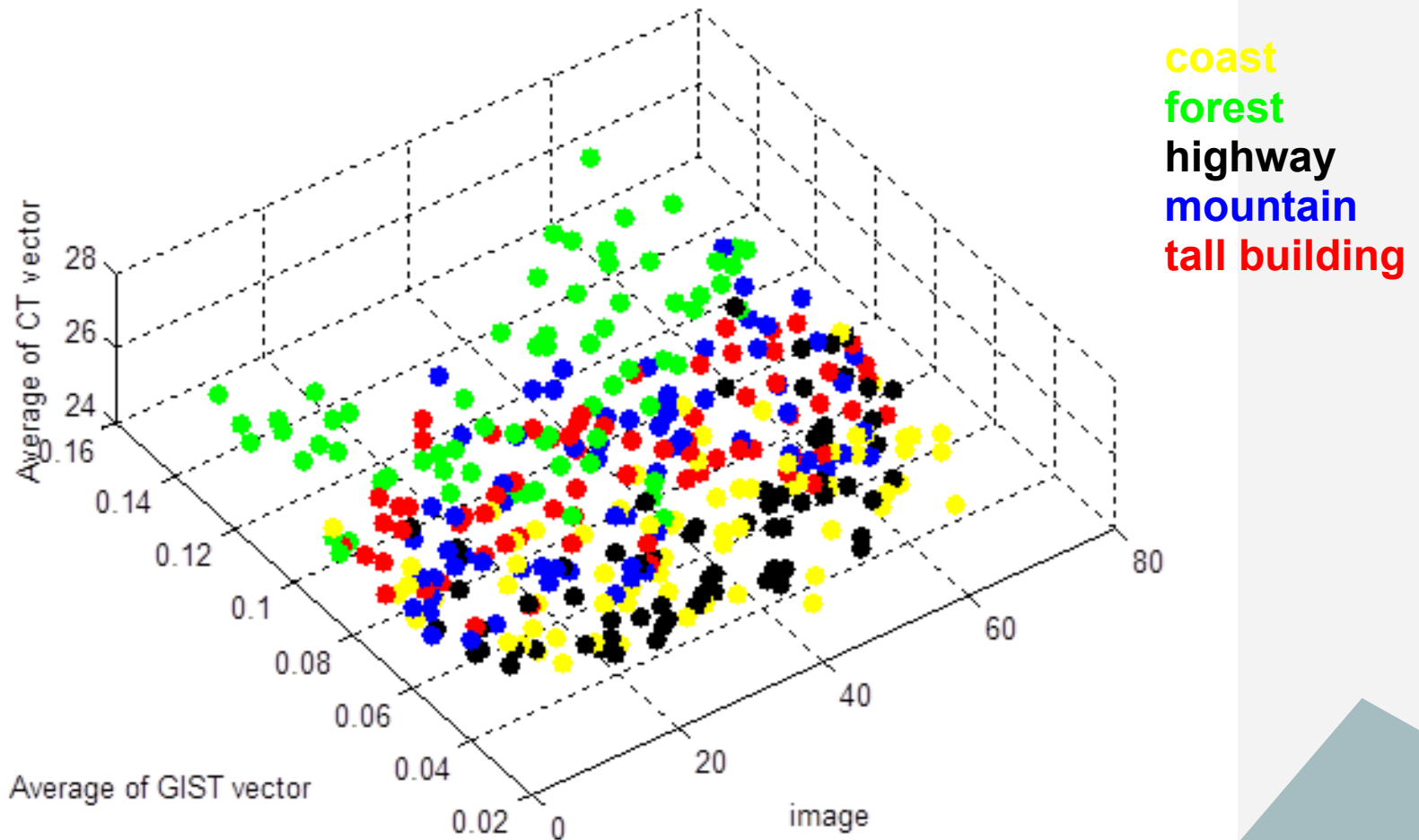
GIST feature vector:



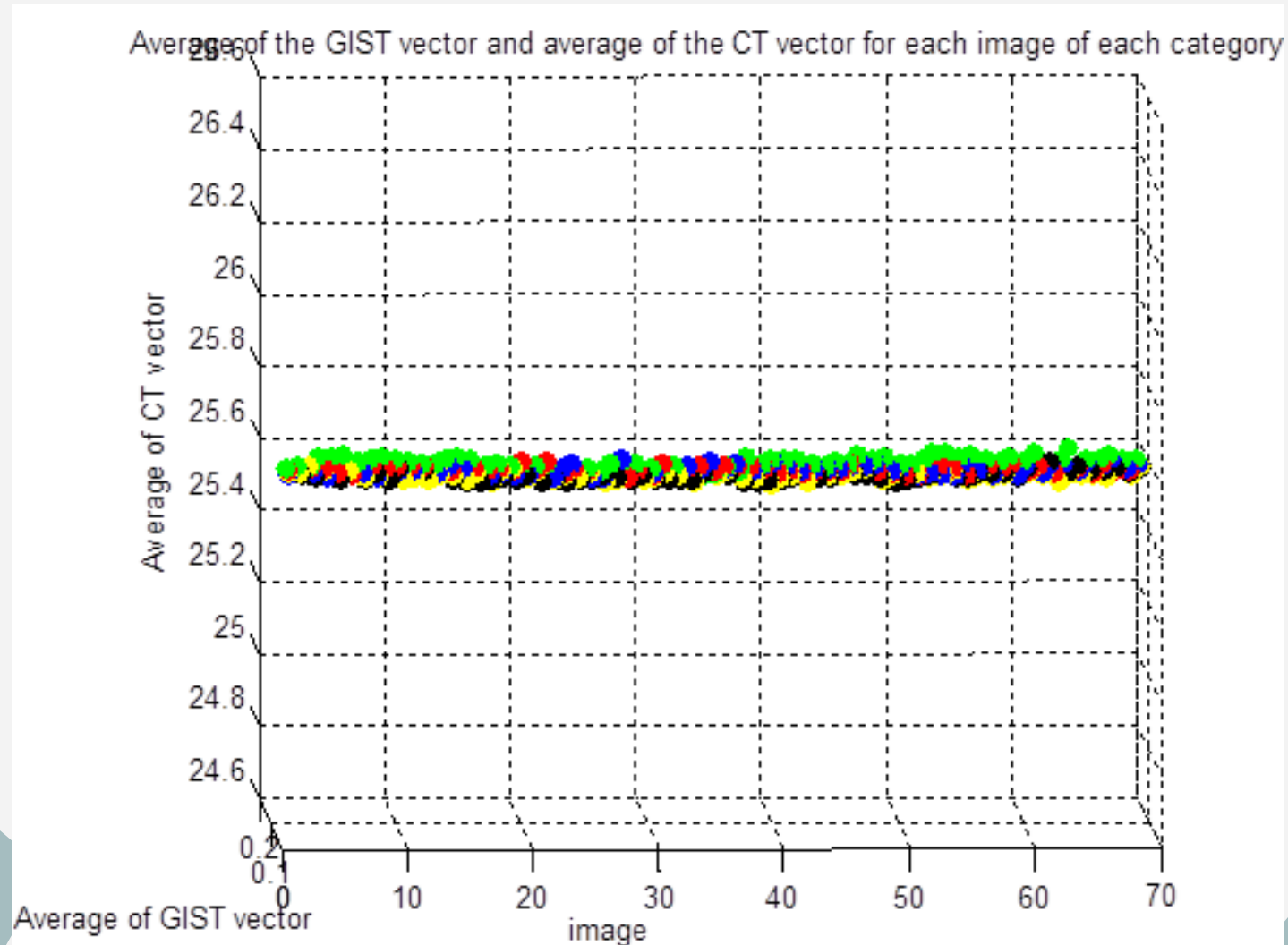
coast
forest
highway
mountain
tall building

Hybrid: Gabor GIST + CENTRIST

Average of the GIST vector and average of the CT vector for each image of each category

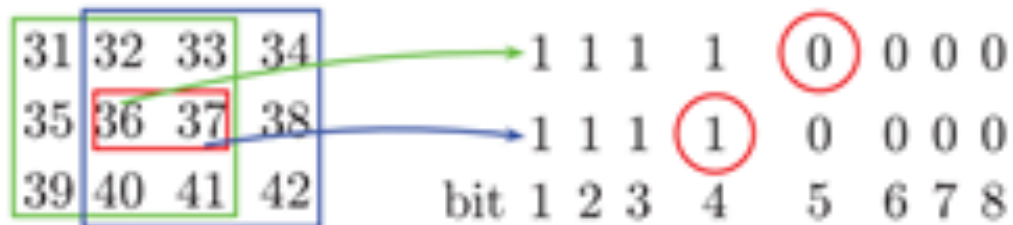


Hybrid: Gabor GIST + CENTRIST



CENTRIST descriptor

32 64 96 1 1 0
32 **64** 96 $\Rightarrow$ 1 0 $\Rightarrow (11010110)_2 \Rightarrow CT = 214$
32 32 96 1 1 0



Constraints between neighbors $\Rightarrow$ homogeneity introduced

CENTRIST descriptor

Pure CENTRIST results through the network:

Training Confusion Matrix

Output Class	1	2	3	4	5	
1	52 21.6%	0 0.0%	4 1.7%	0.4 0.4%	0 0.0%	91.2% 8.8%
2	0 0.0%	43 17.8%	0 0.0%	1 0.4%	1 0.4%	95.6% 4.4%
3	1 0.4%	0 0.0%	42 17.4%	1 0.4%	0 0.0%	95.5% 4.5%
4	0 0.0%	2 0.8%	0 0.0%	38 15.8%	1 0.4%	92.7% 7.3%
5	0 0.0%	2 0.8%	0 0.0%	3 1.2%	49 20.3%	90.7% 9.3%
	98.1% 1.9%	91.5% 8.5%	91.3% 8.7%	86.4% 13.6%	96.1% 3.9%	92.9% 7.1%
Target Class	1	2	3	4	5	

Validation Confusion Matrix

Output Class	1	2	3	4	5	
1	8 15.4%	0 0.0%	2 3.8%	1 1.9%	0 0.0%	72.7% 27.3%
2	0 0.0%	9 17.3%	0 0.0%	2 3.8%	0 0.0%	81.8% 18.2%
3	2 3.8%	0 0.0%	8 15.4%	1 1.9%	0 0.0%	72.7% 27.3%
4	0 0.0%	0 0.0%	1 1.9%	10 19.2%	0 0.0%	90.9% 9.1%
5	0 0.0%	1 1.9%	0 0.0%	0 0.0%	7 13.5%	87.5% 12.5%
	80.0% 20.0%	90.0% 10.0%	72.7% 27.3%	71.4% 28.6%	100% 0.0%	80.8% 19.2%
Target Class	1	2	3	4	5	

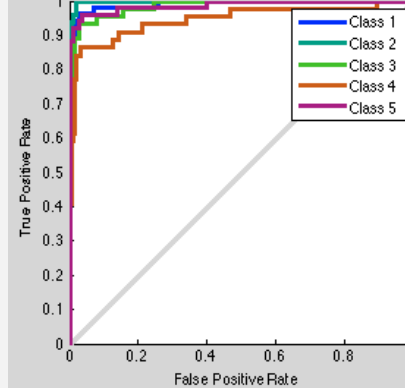
Test Confusion Matrix

Output Class	1	2	3	4	5	
1	6 11.5%	0 0.0%	3 5.8%	1 1.9%	1 1.9%	54.5% 45.5%
2	0 0.0%	12 23.1%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
3	0 0.0%	0 0.0%	7 13.5%	0 0.0%	0 0.0%	100% 0.0%
4	0 0.0%	0 0.0%	0 0.0%	9 17.3%	0 0.0%	100% 0.0%
5	0 0.0%	0 0.0%	2 3.8%	1 1.9%	10 19.2%	76.9% 23.1%
	100% 0.0%	100% 0.0%	58.3% 41.7%	81.8% 18.2%	90.9% 9.1%	84.6% 15.4%
Target Class	1	2	3	4	5	

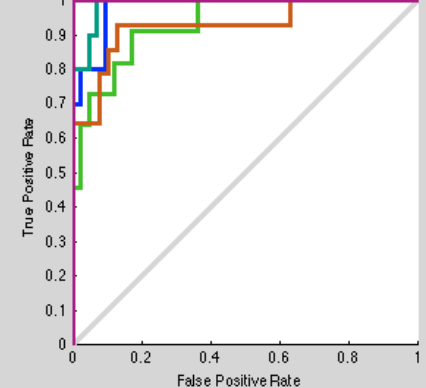
All Confusion Matrix

Output Class	1	2	3	4	5	
1	66 19.1%	0 0.0%	9 2.6%	3 0.9%	1 0.3%	83.5% 16.5%
2	0 0.0%	64 18.6%	0 0.0%	3 0.9%	1 0.3%	94.1% 5.9%
3	3 0.9%	0 0.0%	57 16.5%	2 0.6%	0 0.0%	91.9% 8.1%
4	0 0.0%	2 0.6%	1 0.3%	57 16.5%	1 0.3%	93.4% 6.6%
5	0 0.0%	3 0.9%	2 0.6%	4 1.2%	66 19.1%	88.0% 12.0%
	95.7% 4.3%	92.8% 7.2%	82.6% 17.4%	82.6% 17.4%	95.7% 4.3%	89.9% 10.1%
Target Class	1	2	3	4	5	

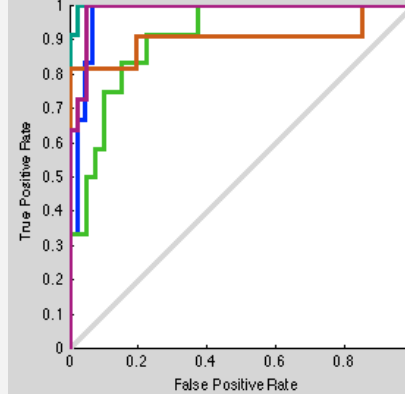
Training ROC



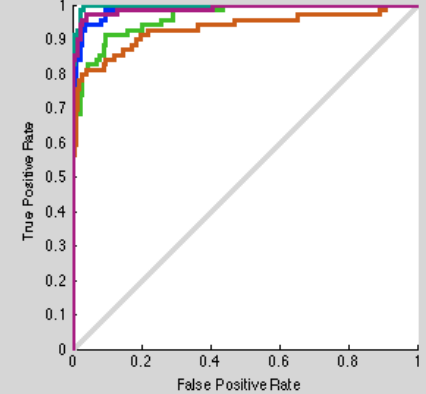
Validation ROC



Test ROC



All ROC



CENTRIST vs Gabor descriptor

Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	
1	23 15.3%	0 0.0%	5 3.3%	0 0.0%	0 0.0%	82.1% 17.9%
2	2 1.3%	28 18.7%	0 0.0%	2 1.3%	0 0.0%	87.5% 12.5%
3	4 2.7%	0 0.0%	25 16.7%	4 2.7%	0 0.0%	75.8% 24.2%
4	1 0.7%	1 0.7%	0 0.0%	23 15.3%	0 0.0%	92.0% 8.0%
5	0 0.0%	1 0.7%	0 0.0%	1 0.7%	30 20.0%	93.8% 6.2%
	76.7% 23.3%	93.3% 6.7%	83.3% 16.7%	76.7% 23.3%	100% 0.0%	86.0% 14.0%



Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	
1	26 17.3%	0 0.0%	4 2.7%	1 0.7%	0 0.0%	83.9% 16.1%
2	1 0.7%	30 20.0%	3 2.0%	0 0.0%	0 0.0%	88.2% 11.8%
3	1 0.7%	0 0.0%	21 14.0%	0 0.0%	1 0.7%	91.3% 8.7%
4	2 1.3%	0 0.0%	2 1.3%	23 15.3%	0 0.0%	85.2% 14.8%
5	0 0.0%	0 0.0%	0 0.0%	6 4.0%	29 19.3%	82.9% 17.1%
	86.7% 13.3%	100% 0.0%	70.0% 30.0%	76.7% 23.3%	96.7% 3.3%	86.0% 14.0%



Hybrid: Gabor GIST + CENTRIST

736 inputs => 200 hidden nodes => 5 outputs

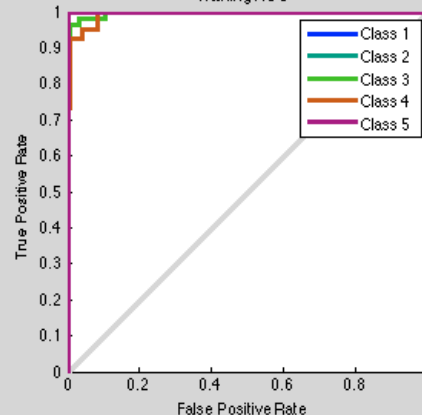
Training Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	Accuracy	Overall Accuracy
1	50 20.7%	0 0.0%	1 0.4%	2 0.8%	0 0.0%	94.3%	5.7%
2	0 0.0%	46 19.1%	0 0.0%	0 0.0%	0 0.0%	100%	0.0%
3	0 0.0%	0 0.0%	52 21.6%	0 0.0%	0 0.0%	100%	0.0%
4	0 0.0%	0 0.0%	0 0.0%	39 16.2%	0 0.0%	100%	0.0%
5	0 0.0%	0 0.0%	0 0.0%	0 0.0%	51 21.2%	100%	0.0%
Overall	100%	100%	98.1%	95.1%	100%	98.8%	1.2%

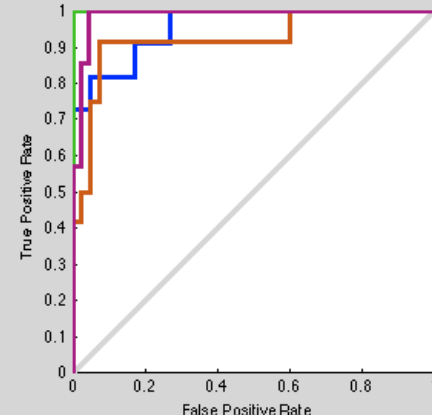
Validation Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	Accuracy	Overall Accuracy
1	8 15.4%	0 0.0%	0 0.0%	1 1.9%	0 0.0%	88.9%	11.1%
2	0 0.0%	14 26.9%	0 0.0%	0 0.0%	0 0.0%	100%	0.0%
3	2 3.8%	0 0.0%	7 13.5%	1 1.9%	0 0.0%	70.0%	30.0%
4	1 1.9%	0 0.0%	0 0.0%	10 19.2%	1 1.9%	83.3%	16.7%
5	0 0.0%	1 1.9%	0 0.0%	0 0.0%	6 11.5%	85.7%	14.3%
Overall	72.7%	93.3%	100%	83.3%	85.7%	86.5%	13.5%

Training ROC



Validation ROC



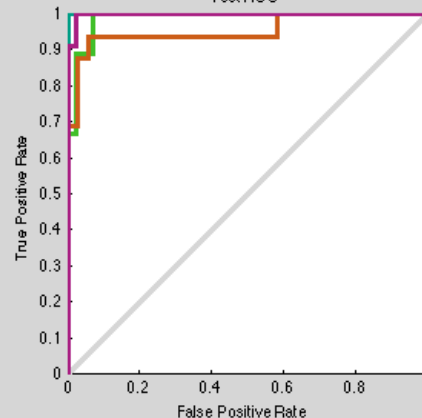
Test Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	Accuracy	Overall Accuracy
1	8 15.4%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100%	0.0%
2	0 0.0%	8 15.4%	0 0.0%	1 1.9%	0 0.0%	88.9%	11.1%
3	0 0.0%	0 0.0%	9 17.3%	1 1.9%	0 0.0%	90.0%	10.0%
4	0 0.0%	0 0.0%	0 0.0%	13 25.0%	0 0.0%	100%	0.0%
5	0 0.0%	0 0.0%	0 0.0%	1 1.9%	11 21.2%	91.7%	8.3%
Overall	100%	100%	100%	81.2%	100%	94.2%	5.8%

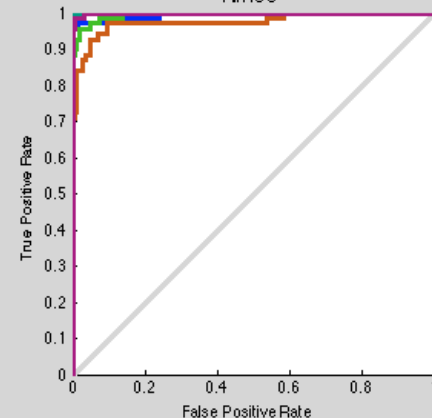
All Confusion Matrix

Output Class \ Target Class	1	2	3	4	5	Accuracy	Overall Accuracy
1	66 19.1%	0 0.0%	1 0.3%	3 0.9%	0 0.0%	94.3%	5.7%
2	0 0.0%	68 19.7%	0 0.0%	1 0.3%	0 0.0%	98.6%	1.4%
3	2 0.6%	0 0.0%	68 19.7%	2 0.6%	0 0.0%	94.4%	5.6%
4	1 0.3%	0 0.0%	0 0.0%	62 18.0%	1 0.3%	96.9%	3.1%
5	0 0.0%	1 0.3%	0 0.0%	1 0.3%	68 19.7%	97.1%	2.9%
Overall	95.7%	98.6%	98.6%	89.9%	98.6%	96.2%	3.8%

Test ROC



All ROC



Hybrid: Gabor GIST + CENTRIST

Confusion Matrix

Output Class	1	2	3	4	5	
	29 19.3%	0 0.0%	4 2.7%	1 0.7%	0 0.0%	85.3% 14.7%
	1 0.7%	30 20.0%	1 0.7%	1 0.7%	1 0.7%	88.2% 11.8%
	0 0.0%	0 0.0%	25 16.7%	3 2.0%	0 0.0%	89.3% 10.7%
	0 0.0%	0 0.0%	0 0.0%	24 16.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	1 0.7%	29 19.3%	96.7% 3.3%
Target Class	1	2	3	4	5	

91.3 % accuracy on 150 testing images dataset..

I have a feeling that it can be boost to 93 or so just by playing with number of hidden neurons and retrain it more and more

Coast and highway got improved comparably with Gabor

Hybrid: Gabor GIST + CENTRIST

Training Confusion Matrix

Output Class	1	2	3	4	5	
	52 21.6%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	47 19.5%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	46 19.1%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	48 19.9%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	48 19.9%	100% 0.0%
	1	2	3	4	5	
	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%	100% 0.0%

Validation Confusion Matrix

Output Class	1	2	3	4	5	
	8 15.4%	0 0.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	8 15.4%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	0 0.0%	0 0.0%	15 28.8%	2 3.8%	0 0.0%	88.2% 11.8%
	0 0.0%	0 0.0%	1 1.9%	8 15.4%	0 0.0%	88.9% 11.1%
	0 0.0%	0 0.0%	0 0.0%	0 0.0%	10 19.2%	100% 0.0%
	1	2	3	4	5	
	100% 0.0%	100% 0.0%	93.8% 6.2%	80.0% 20.0%	100% 0.0%	94.2% 5.8%

Test Confusion Matrix

Output Class	1	2	3	4	5	
	8 15.4%	0 0.0%	0 0.0%	1 1.9%	0 0.0%	88.9% 11.1%
	0 0.0%	13 25.0%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	1 1.9%	0 0.0%	6 11.5%	0 0.0%	0 0.0%	85.7% 14.3%
	0 0.0%	1 1.9%	0 0.0%	10 19.2%	1 1.9%	83.3% 16.7%
	0 0.0%	0 0.0%	1 1.9%	0 0.0%	10 19.2%	90.9% 9.1%
	1	2	3	4	5	
	88.9% 11.1%	92.9% 7.1%	85.7% 14.3%	90.9% 9.1%	90.9% 9.1%	90.4% 9.6%

All Confusion Matrix

Output Class	1	2	3	4	5	
	68 19.7%	0 0.0%	0 0.0%	1 0.3%	0 0.0%	98.6% 1.4%
	0 0.0%	68 19.7%	0 0.0%	0 0.0%	0 0.0%	100% 0.0%
	1 0.3%	0 0.0%	67 19.4%	2 0.6%	0 0.0%	95.7% 4.3%
	0 0.0%	1 0.3%	1 0.3%	66 19.1%	1 0.3%	95.7% 4.3%
	0 0.0%	0 0.0%	1 0.3%	0 0.0%	68 19.7%	98.6% 1.4%
	1	2	3	4	5	
	98.6% 1.4%	98.6% 1.4%	97.1% 2.9%	95.7% 4.3%	98.6% 1.4%	97.7% 2.3%

250 hidden nodes, worse training results, BUT turned to be better overall

Hybrid: Gabor GIST + CENTRIST

Confusion Matrix

Output Class	1	2	3	4	5	
1	28 18.7%	0 0.0%	3 2.0%	1 0.7%	0 0.0%	87.5% 12.5%
2	1 0.7%	30 20.0%	0 0.0%	0 0.0%	0 0.0%	96.8% 3.2%
3	0 0.0%	0 0.0%	25 16.7%	1 0.7%	0 0.0%	96.2% 3.8%
4	1 0.7%	0 0.0%	2 1.3%	27 18.0%	0 0.0%	90.0% 10.0%
5	0 0.0%	0 0.0%	0 0.0%	1 0.7%	30 20.0%	96.8% 3.2%
	93.3% 6.7%	100% 0.0%	83.3% 16.7%	90.0% 10.0%	100% 0.0%	93.3% 6.7%
Target Class	1	2	3	4	5	

93.3 % accuracy on 150 testing images dataset

Improved Mountain and Tall building, but misclassified one coast comparably with 200 nodes.

Now: **100% accuracy** for Forest and Tall building

Gabor descriptor (addition)

About whitening:

Purpose: whitening is simply a process for removing *pairwise, linear* correlations => remove noisy details

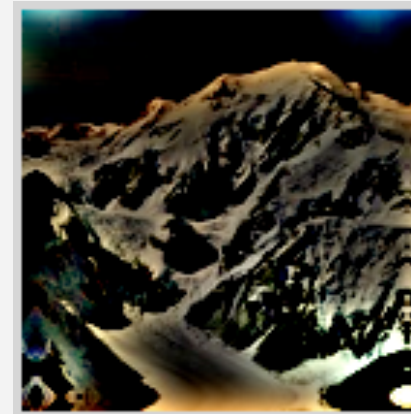
How: through designing a filter in the frequency domain that will flatten the spectrum of a natural image (on average)

- design the amplitude spectrum of the filter so that it rises linearly with frequency, to compensate for the $1/f$ amplitude spectrum of natural images [define a grid of frequency coordinate]

- convolution in the space domain = multiplication in the frequency domain, => multiply the spectrum of the image with the spectrum of the filter and take the inverse fft2

Just check this if interested:

<http://redwood.berkeley.edu/bruno/npb261b/lab2/lab2.html>



The slide features a light gray background. On the left side, there are three overlapping teal-colored geometric shapes that resemble stylized mountain peaks or abstract polygons. Along the bottom edge, there is a continuous teal-colored line composed of several sharp, upward-pointing triangular peaks of varying heights, also resembling a stylized mountain range.

Piter:

Color histogram descriptor

COLOR HISTOGRAM descriptor

